

Cosmology

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1 Introduction

In 1917, Einstein introduces the cosmological constant Λ in his equation

$$G_{\mu\nu} - \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

to ensure his spherical finite universe is static. In 1922, Friedmann derives 2 equations

$$\frac{\dot{a}^2 + kc^2}{a^2} = \frac{8\pi G\rho + \Lambda c^2}{3} \quad \text{and} \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{\Lambda c^2}{3}$$

which show the universe's scale $a(t)$ can change overtime. In 1929, Lemaître and Hubble observe redshift of distant galaxies, which was a good evidence that the universe is expanding. Einstein abandoned the cosmological constant Λ after learning of Hubble's work. In 1998, scientists found that the expansion of the universe is accelerating $\ddot{a} > 0$, which suggested that $\Lambda > 0$.

2 Friedmann-Lemaître-Robertson-Walker metric

The fundamental assumptions for large scales are called the Cosmological principle.

- Homogeneous: The universe has no change under spatial translations.
- Isotropic: The universe has no change under spatial rotations.

There are 3 possible spatially homogeneous and isotropic universe: the flat universe ($k = 0$), the close universe ($k = 1$), and the open universe ($k = -1$).

The flat universe

The flat metric in spherical coordinates is

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -r^2 & 0 \\ 0 & 0 & 0 & -(r \sin \theta)^2 \end{bmatrix}$$

Scaling the spatial part by $\mathbf{e}_i \rightarrow a(t)\mathbf{e}_i$, the spatial metric changes by

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j \rightarrow (a(t))^2 (\mathbf{e}_i \cdot \mathbf{e}_j) = (a(t))^2 g_{ij}$$

Hence, the scaled metric is

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -(a(t))^2 & 0 & 0 \\ 0 & 0 & -(a(t)r)^2 & 0 \\ 0 & 0 & 0 & -(a(t)r \sin \theta)^2 \end{bmatrix}$$

We note that the space time is curved so the Riemann curvature tensor $R^\rho_{\sigma\mu\nu} \neq 0$. However, the spatial part of the metric is flat, so we have $R^k_{lij} = 0$.

The closed universe

For a 3-sphere in a 4-dimensional space

$$\sum_{i=1}^4 x_i^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2 = R^2$$

we can choose (χ, θ, ϕ) as the coordinates on the 3-sphere, using the parametrization

$$x_1 = R \cos \chi$$

$$x_2 = R \sin \chi \cos \theta$$

$$x_3 = R \sin \chi \sin \theta \cos \phi$$

$$x_4 = R \sin \chi \sin \theta \sin \phi$$

The basis vectors are

$$\frac{\partial}{\partial \chi} = \sum_{i=1}^4 \frac{\partial x_i}{\partial \chi} \frac{\partial}{\partial x_i} = -R \sin \chi \frac{\partial}{\partial x_1} + R \cos \chi \cos \theta \frac{\partial}{\partial x_2} + R \cos \chi \sin \theta \cos \phi \frac{\partial}{\partial x_3} + R \cos \chi \sin \theta \sin \phi \frac{\partial}{\partial x_4}$$

$$\frac{\partial}{\partial \theta} = \sum_{i=1}^4 \frac{\partial x_i}{\partial \theta} \frac{\partial}{\partial x_i} = -R \sin \chi \sin \theta \frac{\partial}{\partial x_2} + R \sin \chi \cos \theta \cos \phi \frac{\partial}{\partial x_3} + R \sin \chi \cos \theta \sin \phi \frac{\partial}{\partial x_4}$$

$$\frac{\partial}{\partial \phi} = \sum_{i=1}^4 \frac{\partial x_i}{\partial \phi} \frac{\partial}{\partial x_i} = -R \sin \chi \sin \theta \sin \phi \frac{\partial}{\partial x_3} + R \sin \chi \sin \theta \cos \phi \frac{\partial}{\partial x_4}$$

Using the convention $g_{11} = g_{22} = g_{33} = g_{44} = 1$, the spatial metric is

$$\left\{ \begin{array}{l} g_{\chi\chi} = \frac{\partial}{\partial\chi} \cdot \frac{\partial}{\partial\chi} = R^2 \\ g_{\theta\theta} = \frac{\partial}{\partial\theta} \cdot \frac{\partial}{\partial\theta} = R^2 \sin^2 \chi \\ g_{\phi\phi} = \frac{\partial}{\partial\phi} \cdot \frac{\partial}{\partial\phi} = R^2 \sin^2 \chi \sin^2 \theta \\ g_{\chi\theta} = g_{\chi\phi} = g_{\theta\phi} = 0 \end{array} \right. \Rightarrow g_{ij} = R^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin^2 \chi & 0 \\ 0 & 0 & \sin^2 \chi \sin^2 \theta \end{bmatrix}$$

Absorbing R into $a(t)$, the FLRW metric is

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -(a(t))^2 & 0 & 0 \\ 0 & 0 & -(a(t) \sin \chi)^2 & 0 \\ 0 & 0 & 0 & -(a(t) \sin \chi \sin \theta)^2 \end{bmatrix}$$

Changing the coordinate by $r = \sin \chi$, the metric $g_{\chi\chi}$ is rescaled by

$$\frac{\partial}{\partial r} = \frac{d\chi}{dr} \frac{\partial}{\partial \chi} = \frac{1}{\sqrt{1-r^2}} \frac{\partial}{\partial \chi} \Rightarrow g_{rr} = \frac{\partial}{\partial r} \cdot \frac{\partial}{\partial r} = \frac{g_{\chi\chi}}{1-r^2} = \frac{(a(t))^2}{1-r^2}$$

The FLRW is rewritten by

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{(a(t))^2}{1-r^2} & 0 & 0 \\ 0 & 0 & -(a(t)r)^2 & 0 \\ 0 & 0 & 0 & -(a(t)r \sin \theta)^2 \end{bmatrix}$$

In addition, we can calculate the intrinsic curvature of the 3-sphere. Starting with non-vanishing connecting coefficients

$$\begin{aligned} \Gamma_{\theta\theta}^{\chi} &= -\frac{1}{2} g^{\chi\chi} \partial_{\chi} g_{\theta\theta} = -\sin \chi \cos \chi \\ \Gamma_{\phi\phi}^{\chi} &= -\frac{1}{2} g^{\chi\chi} \partial_{\chi} g_{\phi\phi} = -\sin \chi \cos \chi \sin^2 \theta \\ \Gamma_{\chi\theta}^{\theta} &= \frac{1}{2} g^{\theta\theta} \partial_{\chi} g_{\theta\theta} = \frac{\cos \chi}{\sin \chi} \\ \Gamma_{\phi\phi}^{\theta} &= -\frac{1}{2} g^{\theta\theta} \partial_{\theta} g_{\phi\phi} = -\sin \theta \cos \theta \\ \Gamma_{\chi\phi}^{\phi} &= \frac{1}{2} g^{\phi\phi} \partial_{\chi} g_{\phi\phi} = \frac{\cos \chi}{\sin \chi} \\ \Gamma_{\theta\phi}^{\phi} &= \frac{1}{2} g^{\phi\phi} \partial_{\theta} g_{\phi\phi} = \frac{\cos \theta}{\sin \theta} \end{aligned}$$

Then, the non-vanishing Riemann tensors are

$$\begin{aligned}
R^{\chi}_{\theta\chi\theta} &= \partial_{\chi}\Gamma^{\chi}_{\theta\theta} - \Gamma^{\theta}_{\chi\theta}\Gamma^{\chi}_{\theta\theta} = \sin^2\chi \\
R^{\chi}_{\phi\chi\phi} &= \partial_{\chi}\Gamma^{\chi}_{\phi\phi} - \Gamma^{\phi}_{\chi\phi}\Gamma^{\chi}_{\phi\phi} = \sin^2\chi\sin^2\theta \\
R^{\theta}_{\phi\theta\phi} &= \partial_{\theta}\Gamma^{\theta}_{\phi\phi} + \Gamma^{\chi}_{\phi\phi}\Gamma^{\theta}_{\chi\theta} - \Gamma^{\phi}_{\theta\phi}\Gamma^{\theta}_{\phi\phi} = \sin^2\chi\sin^2\theta \\
R^{\theta}_{\chi\theta\chi} &= g_{\chi\chi}g^{\theta\theta}R^{\chi}_{\theta\chi\theta} = 1 \\
R^{\phi}_{\chi\phi\chi} &= g_{\chi\chi}g^{\phi\phi}R^{\chi}_{\phi\chi\phi} = 1 \\
R^{\phi}_{\theta\phi\theta} &= g_{\theta\theta}g^{\phi\phi}R^{\theta}_{\phi\theta\phi} = \sin^2\chi
\end{aligned}$$

The non-vanishing Ricci tensors are

$$\begin{aligned}
R_{\chi\chi} &= R^{\theta}_{\chi\theta\chi} + R^{\phi}_{\chi\phi\chi} = 2 \\
R_{\theta\theta} &= R^{\chi}_{\theta\chi\theta} + R^{\phi}_{\theta\phi\theta} = 2\sin^2\chi \\
R_{\phi\phi} &= R^{\chi}_{\phi\chi\phi} + R^{\theta}_{\phi\theta\phi} = 2\sin^2\chi\sin^2\theta
\end{aligned}$$

The Ricci scalar is then

$$\mathcal{R} = g^{\chi\chi}R_{\chi\chi} + g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{6}{R^2}$$

The open universe

For a hyperbolic 3-space in a 4-dimensional space

$$-x_1^2 + x_2^2 + x_3^2 + x_4^2 = -R^2$$

we can also parametrize the space by (χ, θ, ϕ)

$$x_1 = R \cosh \chi$$

$$x_2 = R \sinh \chi \cos \theta$$

$$x_3 = R \sinh \chi \sin \theta \cos \phi$$

$$x_4 = R \sinh \chi \sin \theta \sin \phi$$

The basis vectors are

$$\begin{aligned}\frac{\partial}{\partial \chi} &= R \sinh \chi \frac{\partial}{\partial x_1} + R \cosh \chi \cos \theta \frac{\partial}{\partial x_2} + R \cosh \chi \sin \theta \cos \phi \frac{\partial}{\partial x_3} + R \cosh \chi \sin \theta \sin \phi \frac{\partial}{\partial x_4} \\ \frac{\partial}{\partial \theta} &= -R \cosh \chi \sin \theta \frac{\partial}{\partial x_2} + R \cosh \chi \cos \theta \cos \phi \frac{\partial}{\partial x_3} + R \cosh \chi \cos \theta \sin \phi \frac{\partial}{\partial x_4} \\ \frac{\partial}{\partial \phi} &= -R \cosh \chi \sin \theta \sin \phi \frac{\partial}{\partial x_3} + R \cosh \chi \sin \theta \cos \phi \frac{\partial}{\partial x_4}\end{aligned}$$

Using the convention $-g_{11} = g_{22} = g_{33} = g_{44} = 1$, the spatial metric is

$$\left\{ \begin{aligned} g_{\chi\chi} &= \frac{\partial}{\partial \chi} \cdot \frac{\partial}{\partial \chi} = R^2 \\ g_{\theta\theta} &= \frac{\partial}{\partial \theta} \cdot \frac{\partial}{\partial \theta} = R^2 \sinh^2 \chi \\ g_{\phi\phi} &= \frac{\partial}{\partial \phi} \cdot \frac{\partial}{\partial \phi} = R^2 \sinh^2 \chi \sin^2 \theta \\ g_{\chi\theta} &= g_{\chi\phi} = g_{\theta\phi} = 0 \end{aligned} \right. \Rightarrow g_{ij} = R^2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \sinh^2 \chi & 0 & 0 \\ 0 & 0 & \sinh^2 \chi \sin^2 \theta & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If we calculate the curvature of the 3-space, we will see $\mathcal{R} = -\frac{6}{R^2}$.

Absorbing R into $a(t)$, the FLRW metric is then

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -(a(t))^2 & 0 & 0 \\ 0 & 0 & -(a(t) \sinh \chi)^2 & 0 \\ 0 & 0 & 0 & -(a(t) \sinh \chi \sin \theta)^2 \end{bmatrix}$$

Changing the coordinate by $r = \sinh \chi$, the metric $g_{\chi\chi}$ is rescaled by

$$\frac{\partial}{\partial r} = \frac{d\chi}{dr} \frac{\partial}{\partial \chi} = \frac{1}{\sqrt{1+r^2}} \frac{\partial}{\partial \chi} \Rightarrow g_{rr} = \frac{\partial}{\partial r} \cdot \frac{\partial}{\partial r} = \frac{g_{\chi\chi}}{1+r^2} = \frac{(a(t))^2}{1+r^2}$$

The FLRW is rewritten by

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{(a(t))^2}{1+r^2} & 0 & 0 \\ 0 & 0 & -(a(t)r)^2 & 0 \\ 0 & 0 & 0 & -(a(t)r \sin \theta)^2 \end{bmatrix}$$

In conclusion, all three cases can be expressed by a parameter k

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{(a(t))^2}{1-kr^2} & 0 & 0 \\ 0 & 0 & -(a(t)r)^2 & 0 \\ 0 & 0 & 0 & -(a(t)r \sin \theta)^2 \end{bmatrix}, \quad \text{where } k = \begin{cases} 1, & \text{closed universe} \\ 0, & \text{flat universe} \\ -1, & \text{open universe} \end{cases}$$

3 Curvature tensor of FLRW metric

To simplify the calculations, we define

$$q_{ij} = \begin{bmatrix} \frac{1}{1-kr^2} & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}, \quad i, j = 1, 2, 3$$

First, we calculate the non-vanishing connecting coefficients

$$\begin{aligned} \Gamma_{ii}^0 &= -\frac{1}{2}g^{00}\partial_0 g_{ii} = -\frac{1}{2c}\frac{\partial}{\partial t}(-a^2 q_{ii}) = \frac{a\dot{a}}{c}q_{ii} \\ \Gamma_{0i}^i &= \frac{1}{2}g^{ii}\partial_0 g_{ii} = \frac{1}{2c(-a^2 q_{ii})}\frac{\partial}{\partial t}(-a^2 q_{ii}) = \frac{\dot{a}}{ca} \\ \Gamma_{11}^1 &= \frac{1}{2}g^{11}\partial_1 g_{11} = \frac{kr}{1-kr^2} \\ \Gamma_{22}^1 &= -\frac{1}{2}g^{11}\partial_1 g_{22} = -r(1-kr^2) \\ \Gamma_{33}^1 &= -\frac{1}{2}g^{11}\partial_1 g_{33} = -r(1-kr^2)\sin^2 \theta \\ \Gamma_{33}^2 &= -\frac{1}{2}g^{22}\partial_2 g_{33} = -\sin \theta \cos \theta \\ \Gamma_{12}^2 &= \frac{1}{2}g^{22}\partial_1 g_{22} = \frac{1}{r} \\ \Gamma_{13}^3 &= \frac{1}{2}g^{33}\partial_1 g_{33} = \frac{1}{r} \\ \Gamma_{23}^3 &= \frac{1}{2}g^{33}\partial_2 g_{33} = \frac{\cos \theta}{\sin \theta} \end{aligned}$$

Then, the non-vanishing Ricci tensor

$$\begin{aligned} R_{00} &= R^\mu_{0\mu 0} = -\partial_0 \Gamma_{0\mu}^\mu - \Gamma_{0\mu}^\alpha \Gamma_{\alpha 0}^\mu = -\frac{3(\ddot{a}a - \dot{a}^2)}{c^2 a^2} - \frac{3\dot{a}^2}{c^2 a^2} = -\frac{3\ddot{a}}{c^2 a} \\ R_{11} &= R^\mu_{1\mu 1} = \partial_\mu \Gamma_{11}^\mu - \partial_1 \Gamma_{1\mu}^\mu + \Gamma_{11}^\alpha \Gamma_{\alpha\mu}^\mu - \Gamma_{1\mu}^\alpha \Gamma_{\alpha 1}^\mu \\ &= \partial_0 \Gamma_{11}^0 - 2\partial_1 \Gamma_{12}^2 + \Gamma_{11}^0 \Gamma_{01}^1 + 2\Gamma_{11}^1 \Gamma_{12}^2 - 2\Gamma_{12}^2 \Gamma_{21}^2 \\ &= \frac{\ddot{a}a + \dot{a}^2}{c^2(1-kr^2)} + \frac{2}{r^2} + \frac{\dot{a}^2}{c^2(1-kr^2)} + \frac{2k}{1-kr^2} - \frac{2}{r^2} \\ &= \frac{\ddot{a}a + 2\dot{a}^2 + 2kc^2}{c^2(1-kr^2)} \end{aligned}$$

$$\begin{aligned}
R_{22} &= R^\mu{}_{2\mu 2} = \partial_\mu \Gamma_{22}^\mu - \partial_2 \Gamma_{2\mu}^\mu + \Gamma_{22}^\alpha \Gamma_{\alpha\mu}^\mu - \Gamma_{2\mu}^\alpha \Gamma_{\alpha 2}^\mu \\
&= \partial_0 \Gamma_{22}^0 + \partial_1 \Gamma_{22}^1 - \partial_2 \Gamma_{23}^3 + \Gamma_{22}^0 \Gamma_{01}^1 + \Gamma_{22}^1 \Gamma_{11}^1 - \Gamma_{23}^3 \Gamma_{32}^3 \\
&= \frac{a\ddot{a} + \dot{a}^2}{c^2} r^2 - (1 - 3kr^2) + \frac{1}{\sin^2 \theta} + \frac{\dot{a}^2}{c^2} r^2 - kr^2 - \frac{\cos^2 \theta}{\sin^2 \theta} \\
&= \frac{a\ddot{a} + 2\dot{a}^2 + 2kc^2}{c^2} r^2
\end{aligned}$$

$$\begin{aligned}
R_{33} &= R^\mu{}_{3\mu 3} = \partial_\mu \Gamma_{33}^\mu - \partial_3 \Gamma_{3\mu}^\mu + \Gamma_{33}^\alpha \Gamma_{\alpha\mu}^\mu - \Gamma_{3\mu}^\alpha \Gamma_{\alpha 3}^\mu \\
&= \partial_0 \Gamma_{33}^0 + \partial_1 \Gamma_{33}^1 + \partial_2 \Gamma_{33}^2 + \Gamma_{33}^0 \Gamma_{01}^1 + \Gamma_{33}^1 \Gamma_{11}^1 - \Gamma_{33}^2 \Gamma_{23}^3 \\
&= \frac{a\ddot{a} + \dot{a}^2}{c^2} r^2 \sin^2 \theta - (1 - 3kr^2) \sin^2 \theta - \cos^2 \theta + \sin^2 \theta + \frac{\dot{a}^2}{c^2} r^2 \sin^2 \theta - kr^2 \sin^2 \theta + \cos^2 \theta \\
&= \frac{a\ddot{a} + 2\dot{a}^2 + 2kc^2}{c^2} r^2 \sin^2 \theta
\end{aligned}$$

We can summarize the tensor by

$$R_{ii} = \frac{a\ddot{a} + 2\dot{a}^2 + 2kc^2}{c^2} q_{ii}$$

Then, the Ricci scalar is

$$R = g^{\mu\nu} R_{\mu\nu} = -\frac{3\ddot{a}}{c^2 a} - 3 \left(\frac{a\ddot{a} + 2\dot{a}^2 + 2kc^2}{c^2 a^2} \right) = -6 \left(\frac{a\ddot{a} + \dot{a}^2 + kc^2}{c^2 a^2} \right)$$

4 FLRW geodesics and comoving coordinates

The geodesics satisfy the geodesic equation

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0$$

Comoving coordinates

It is trivial that $x^\mu(\lambda) = (ct_0 + \lambda, r_0, \theta_0, \phi_0)$ is a geodesic because

$$\frac{dx^\mu}{d\lambda} = (1, 0, 0, 0) \quad \text{and} \quad \frac{d^2 x^\mu}{d\lambda^2} = (0, 0, 0, 0)$$

and then we only need $\Gamma_{00}^\mu = 0$ for $\mu = 0, 1, 2, 3$, which is correctly satisfied. The coordinates

$$x^\mu(\lambda) = (ct_0 + \lambda, r_0, \theta_0, \phi_0)$$

are called the comoving coordinates since the spatial part is constant. However, the distance between two spatial coordinates is not constant. For the distance along radial direction

$$r = \lambda \quad \Rightarrow \quad dr = d\lambda, \quad dt = d\theta = d\phi = 0$$

we can integrate

$$L_0 = \int \frac{ds}{d\lambda} d\lambda = \int_0^r \sqrt{-g_{rr} \left(\frac{dr}{d\lambda} \right)^2} d\lambda = \int_0^r \frac{a(t)}{\sqrt{1 - k\lambda^2}} d\lambda = \begin{cases} a(t) \sin^{-1} r, & \text{for } k = 1 \\ a(t)r, & \text{for } k = 0 \\ a(t) \sinh^{-1} r, & \text{for } k = -1 \end{cases}$$

Therefore, the distance between two free particles changes overtime, when the FLRW metric has a changing scale factor $a(t)$.

Light-like geodesics

For radial light-like geodesics $d\theta = d\phi = 0$, we solve the condition

$$0 = c^2 d\tau^2 = c^2 dt^2 - \frac{a(t)^2}{1 - kr^2} dr^2 \quad \Rightarrow \quad \frac{dr}{dt} = \frac{c}{a(t)} \sqrt{1 - kr^2}$$

For the flat space time $k = 0$, we see $\frac{dr}{dt} = \frac{c}{a(t)}$, which shows that all radial lights have the same coordinate speed at a given time t . Also, we can consider a light beam emitted at t_0 with frequency f_0 and then received at t with frequency f . Since the coordinate distance of two wave peaks of light is a constant Δr , the wavelength is calculated by

$$\begin{cases} \lambda_{\text{emit}} = a(t_0) \Delta r \\ \lambda_{\text{receive}} = a(t) \Delta r \end{cases} \quad \Rightarrow \quad \frac{\lambda_{\text{receive}}}{\lambda_{\text{emit}}} = \frac{f_0}{f} = \frac{a(t)}{a(t_0)} = 1 + z$$

For $z > 0$, $f_0 > f$, $a(t_0) < a(t)$, saying that redshift occurs in an expanding universe. Now, one may ask if star systems, people, atoms also expand along with universe. The answer is negative. Particles that make up atoms are affected by electromagnetic forces and strong nuclear forces, so they are not free particles. For star systems and individual galaxies, the objects inside these systems are essentially free particles only affected by the curvature of the space time, though, they still do not expand with the universe. FLRW metric assumes that the universe is homogeneous and isotropic, so FLRW metric does not apply to small scales described by Schwarzschild metric involving smaller bodies orbiting large central mass.

Cosmological horizons

There are three horizons to be discussed.

Hubble horizon

The Hubble horizon is the boundary beyond the galaxies that are travelling away faster than light. The velocities of these galaxies today are described by the Hubble constant $H_0 \approx 68 \text{ km/s per Mpc}$. So the Hubble horizon is given by

$$L_{\text{HH}}(t) = \frac{c}{H(t)}$$

where $H(t)$ is the Hubble parameter varies with time.

Event horizon (Cosmic horizon)

The event horizon is the maximum comoving distance a light beam emitted from Earth can travel. In flat space time, the event horizon is given by

$$\frac{dr}{dt} = \frac{c}{a(t)} \quad \Rightarrow \quad r_{\text{EH}}(t) = \int_{t_{\text{now}}}^{\infty} \frac{c}{a(t)} dt$$

The proper length of the event horizon is

$$L_{\text{EH}}(t) = a(t) \int_{t_{\text{now}}}^{\infty} \frac{c}{a(t')} dt'$$

If the integral is infinite, then there is no event horizon, and light can reach any point in the universe. Here, we can discuss that if the Hubble parameter $H(t)$ is constant, then the Hubble horizon and the event horizon are the same.

Particle horizon

The particle horizon is the distance a light beam has travelled since the beginning of the universe. In flat space time, the particle horizon is given by

$$\frac{dr}{dt} = \frac{c}{a(t)} \quad \Rightarrow \quad r_{\text{PH}}(t) = \int_0^{t_{\text{now}}} \frac{c}{a(t)} dt \quad \Rightarrow \quad L_{\text{PH}}(t) = a(t) \int_0^{t_{\text{now}}} \frac{c}{a(t')} dt'$$

We can see in comoving coordinates r_{EH} is always shrinking and r_{PH} is always growing overtime.

5 Universe as a perfect fluid

Energy-stress tensor of perfect fluid

The energy-stress tensor $T^{\mu\nu}$ has 16 components including

- T^{00} : the energy density
- T^{0i} : the i^{th} -component momentum density
- T^{i0} : the i^{th} -component flux of the energy density
- T^{ij} : the j^{th} -component flux of the i^{th} -component momentum density

The momentum flux T^{ij} can also be separated into

- the pressure T^{11}, T^{22}, T^{33}
- the shear stress T^{12}, T^{13}, T^{23}

Since the energy-stress tensor is symmetric, we have $T^{0i} = T^{i0}$ and $T^{ij} = T^{ji}$. In cosmic rest frame, the isotropy of the universe require zero energy flux, zero shear stress, and the same pressure in all direction.

$$T^{i0} = T^{12} = T^{13} = T^{23} = 0, \quad T^{11} = T^{22} = T^{33}$$

Then, the energy tensor in Minkowski space can be parametrized by

$$T^{\mu\nu} = \begin{bmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{bmatrix}$$

We can also separate $T^{\mu\nu}$ into two tensors, using a time-like velocity vector $u^\mu = (c, 0, 0, 0)$.

$$T^{\mu\nu} = \left(\rho + \frac{p}{c^2} \right) u^\mu u^\nu - p \eta^{\mu\nu}$$

For FLRW metric, we just change $\eta^{\mu\nu}$ to $g^{\mu\nu}$, so the energy tensor is

$$T_{\mu\nu} = \left(\rho + \frac{p}{c^2} \right) u_\mu u_\nu - p g_{\mu\nu} = \begin{bmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & \frac{pa^2}{1-kr^2} & 0 & 0 \\ 0 & 0 & pa^2 r^2 & 0 \\ 0 & 0 & 0 & pa^2 r^2 \sin^2 \theta \end{bmatrix}$$

It is cleaner if we only lower one index of $T^{\mu\nu}$.

$$T^{\mu}_{\nu} = \left(\rho + \frac{p}{c^2} \right) u^{\mu} u_{\nu} - p \delta^{\mu}_{\nu} = \begin{bmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & -p & 0 & 0 \\ 0 & 0 & -p & 0 \\ 0 & 0 & 0 & -p \end{bmatrix}$$

Cosmic rest frame

One may ask why there is a rest frame for our universe since there should not be an objective rest frame, or even an objective t coordinate in SR. The reason is that we define the cosmic rest frame as the unique frame where

- there is no Doppler shift in the CMB radiation
- the average velocity of masses at large scales is zero

Hence, if the universe is empty, there is no CMB radiation for us to define the cosmic rest frame, and the space time itself becomes homogeneous and isotropic. Here, we can classify

- The empty space time with zero curvature is the Minkowski space
- The empty space time with positive curvature is the de Sitter space
- The empty space time with negative curvature is the Anti-de Sitter space

Equation of states

As in thermodynamics, the equation of states relates the states variables together. We discuss the simplest option

$$p = w \rho c^2$$

where w is a pure number. This equation describes the universe as barotropic fluid, where $\rho(p)$ is a function of pressure only.

- If we set $w = 0$, $p = 0$, we obtain the stress energy tensor with matter only.

$$T^{\mu}_{\nu} = \begin{bmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- If we set $w = \frac{1}{3}$, $\rho c^2 = 3p$, we obtain traceless stress energy tensor $T^\mu{}_\mu = 0$.

$$T^\mu{}_\nu = \begin{bmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & -\frac{1}{3}\rho c^2 & 0 & 0 \\ 0 & 0 & -\frac{1}{3}\rho c^2 & 0 \\ 0 & 0 & 0 & -\frac{1}{3}\rho c^2 \end{bmatrix}$$

which means that there is only radiation.

- If we set $w = -1$, $p = -\rho c^2$, the energy tensor can be simplified by

$$T^\mu{}_\nu = \rho c^2 \delta^\mu{}_\nu \quad \Rightarrow \quad T_{\mu\nu} = \rho c^2 g_{\mu\nu}$$

If we let $\Lambda = \rho c^2$, the Einstein equation becomes

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R - \rho c^2 g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \quad \Rightarrow \quad R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \tilde{T}_{\mu\nu} + \frac{8\pi G}{c^4}T_{\mu\nu}$$

where we also let $\tilde{T}_{\mu\nu} = \rho c^2 g_{\mu\nu}$. We will later see the negative pressure $p = -\rho c^2$ represents the dark energy, which causes the universe expansion to accelerate outward.

6 Friedmann equations

The 1st Friedmann equation is the 00-component of the Einstein equation.

$$\begin{aligned} R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} - \Lambda g_{\mu\nu} &= \frac{8\pi G}{c^4}T_{\mu\nu} \quad \Rightarrow \quad -\frac{3\ddot{a}}{c^2 a} + 3\left(\frac{a\ddot{a} + \dot{a}^2 + kc^2}{c^2 a^2}\right) - \Lambda = \frac{8\pi G}{c^2}\rho \\ &\Rightarrow \quad \frac{\dot{a}^2 + kc^2}{a^2} = \frac{8\pi G\rho + \Lambda c^2}{3} \end{aligned}$$

The 2nd Friedmann equation is the trace of the Einstein equation.

$$\begin{aligned} g^{\mu\nu} \left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} - \Lambda g_{\mu\nu} \right) &= \frac{8\pi G}{c^4}g^{\mu\nu}T_{\mu\nu} \quad \Rightarrow \quad R - 2R - 4\Lambda = \frac{8\pi G}{c^4}(\rho c^2 - 3p) \\ &\Rightarrow \quad \frac{\ddot{a}}{a} + \frac{\dot{a}^2 + kc^2}{a^2} = \frac{4\pi G}{3} \left(\rho - \frac{3p}{c^2} \right) + \frac{2}{3}\Lambda c^2 \end{aligned}$$

Eliminate $\frac{\dot{a}^2 + kc^2}{a^2}$ by the 1st Friedmann equation, we obtain

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{1}{3}\Lambda c^2$$

If we take the derivative of the 1st Friedmann equation, and eliminate $\ddot{a}$ and $\frac{\dot{a}^2 + kc^2}{a^2}$, we have

$$\begin{aligned}\frac{8\pi G}{3}\dot{\rho} &= \frac{2\dot{a}\ddot{a}}{a^2} - \frac{2\dot{a}(\dot{a}^2 + kc^2)}{a^3} \\ &= \frac{2\dot{a}}{a} \left(\frac{\ddot{a}}{a} - \frac{\dot{a}^2 + kc^2}{a^2} \right) \\ &= \frac{2\dot{a}}{a} \left[-\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{1}{3}\Lambda c^2 - \frac{8\pi G\rho + \Lambda c^2}{3} \right] \\ &= -8\pi G \left(\rho + \frac{p}{c^2} \right) \frac{\dot{a}}{a}\end{aligned}$$

Eliminating $8\pi G$, we obtain the fluid equation.

$$\dot{\rho} = -\frac{3\dot{a}}{a} \left(\rho + \frac{p}{c^2} \right)$$

This equation can be derived in other two ways.

(a) Considering the expansion of the universe is adiabatic, we apply the 1st law of thermodynamics.

$$\begin{aligned}0 &= dQ = dE + p dV \\ &= d(\rho c^2 a^3) + p d(a^3) \\ &= c^2 a^3 d\rho + 3\rho c^2 a^2 da + 3pa^2 da\end{aligned}$$

Dividing the equation by dt , we obtain

$$\dot{\rho} + \frac{3\dot{a}}{a} \left(\rho + \frac{p}{c^2} \right) = 0$$

(b) The local conservation of energy-momentum gives

$$0 = \nabla_\mu T^\mu{}_\nu = \partial_\mu T^\mu{}_\nu + \Gamma_{\mu\alpha}^\mu T^\alpha{}_\nu - \Gamma_{\mu\nu}^\alpha T^\mu{}_\alpha$$

The $\mu = 0$ component gives

$$\begin{aligned}0 &= \partial_\mu T^\mu{}_0 + \Gamma_{\mu\alpha}^\mu T^\alpha{}_0 - \Gamma_{\mu 0}^\alpha T^\mu{}_\alpha \\ &= \partial_0 T^0{}_0 + 3\Gamma_{10}^1 T^0{}_0 - 3\Gamma_{10}^1 T^1{}_1 \\ &= \dot{\rho}c + \frac{3\dot{a}}{a}\rho c + \frac{3\dot{a}}{ac}p \quad \Rightarrow \quad \dot{\rho} + \frac{3\dot{a}}{a} \left(\rho + \frac{p}{c^2} \right) = 0\end{aligned}$$

Matter, radiation, and dark matter

We substitute the equation of states into the fluid equation.

$$\dot{\rho} = -\frac{3\dot{a}}{a}(\rho + w\rho) \Rightarrow \frac{\dot{\rho}}{\rho} = -3(1+w)\frac{\dot{a}}{a} \Rightarrow \rho(a) = Ka^{-3(1+w)}$$

- For matter dominated universe $w = 0$, we have $\rho(a) \propto a^{-3}$. It shows that the mass density changes inversely with the scale factor in all three spatial dimensions.
- For radiation dominated universe $w = \frac{1}{3}$, we have $\rho(a) \propto a^{-4}$. The extra scale factor comes from the cosmological redshift.

$$E = \frac{hc}{\lambda} = \frac{hc}{\lambda_e} \frac{a_e}{a} \propto a^{-1}$$

- For dark energy dominated universe $w = -1$, we have $\rho(a) \propto 1$. It shows that a constant volume always has the same constant amount of dark energy. Increasing the size of the universe fundamentally increases the amount of dark energy.

For the cases $w = \frac{1}{3}$ and $w = -1$, we see that the energy is not globally conserved in General Relativity. According to Noether's Theorem, the conservation of energy holds only if the universe is invariant under time translation. However, the scalar factor $a(t)$ changes overtime, so we do not have time-translation symmetry. Basicly, the energy is conserved locally $\nabla_\mu T^{\mu\nu} = 0$ at any point, but we can not extend this globally.

7 Universe evolution models

Einstein static universe

If the universe is not expanding, we set $\dot{a} = \ddot{a} = 0$, then we can solve

$$\begin{aligned} \frac{\ddot{a}}{a} = 0 &= -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{1}{3} \Lambda c^2 \Rightarrow \Lambda = \frac{4\pi G}{c^2} \left(\rho + \frac{3p}{c^2} \right) \\ \frac{\dot{a}^2 + kc^2}{a^2} &= \frac{kc^2}{a^2} = \frac{8\pi G\rho + \Lambda c^2}{3} \Rightarrow \frac{k}{a^2} = \frac{4\pi G}{c^2} \left(\rho + \frac{p}{c^2} \right) \end{aligned}$$

Einstein also assumes zero pressure $p = 0$, so we obtain

$$\frac{k}{a^2} = \Lambda = \frac{4\pi G}{c^2} \rho$$

Since $\rho > 0$, the universe has to be spherical $k = 1$, and the cosmological constant is the inverse radius squared, which is also positive.

$$\Lambda = \frac{1}{a^2} > 0$$

This model can not describe our expanding universe.

De Sitter universe

If there is no matter and radiation $\rho = p = 0$, the 1st Friedmann equation becomes

$$\frac{\dot{a}^2 + kc^2}{a^2} = \frac{\Lambda c^2}{3}$$

- For flat universe $k = 0$, we solve

$$\dot{a} = a\sqrt{\frac{\Lambda c^2}{3}} \Rightarrow a(t) = e^{\sqrt{\frac{\Lambda c^2}{3}}t}$$

Calling $R = \sqrt{\frac{3}{\Lambda}}$, we can embed the metric

$$ds^2 = -c^2 dt^2 + e^{\frac{ct}{R}} (d\chi^2 + \chi^2 d\theta^2 + \chi^2 \sin^2 \theta d\phi^2)$$

using the parametrization of a hyperboloid.

$$\left\{ \begin{array}{l} x_0 = 2R \sinh\left(\frac{ct}{2R}\right) + \frac{\chi^2}{4R} e^{\frac{ct}{2R}} \\ x_1 = 2R \cosh\left(\frac{ct}{2R}\right) - \frac{\chi^2}{4R} e^{\frac{ct}{2R}} \\ x_2 = e^{\frac{ct}{2R}} \chi \cos \theta \\ x_3 = e^{\frac{ct}{2R}} \chi \sin \theta \cos \phi \\ x_4 = e^{\frac{ct}{2R}} \chi \sin \theta \sin \phi \end{array} \right. \Rightarrow -x_0^2 + \sum_{i=1}^4 x_i^2 = R^2$$

- For closed universe $k = 1$, we solve

$$\dot{a}^2 = \frac{\Lambda c^2}{3} a^2 - c^2 \Rightarrow a(t) = \sqrt{\frac{3}{\Lambda}} \cosh\left(\sqrt{\frac{\Lambda c^2}{3}}t\right)$$

Calling $R = \sqrt{\frac{3}{\Lambda}}$, we can embed the metric

$$ds^2 = -c^2 dt^2 + R^2 \cosh^2\left(\frac{ct}{R}\right) (d\chi^2 + \sin^2 \chi d\theta^2 + \sin^2 \chi \sin^2 \theta d\phi^2)$$

using the parametrization of a hyperboloid.

$$\begin{cases} x_0 = R \sinh \left(\frac{ct}{R} \right) \\ x_1 = R \cosh \left(\frac{ct}{R} \right) \cos \chi \\ x_2 = R \cosh \left(\frac{ct}{R} \right) \sin \chi \cos \theta \\ x_3 = R \cosh \left(\frac{ct}{R} \right) \sin \chi \sin \theta \cos \phi \\ x_4 = R \cosh \left(\frac{ct}{R} \right) \sin \chi \sin \theta \sin \phi \end{cases} \Rightarrow -x_0^2 + \sum_{i=1}^4 x_i^2 = R^2$$

- For open universe $k = -1$, we solve

$$\dot{a}^2 = \frac{\Lambda c^2}{3} a^2 + c^2 \Rightarrow a(t) = \sqrt{\frac{3}{\Lambda}} \sinh \left(\sqrt{\frac{\Lambda c^2}{3}} t \right)$$

Calling $R = \sqrt{\frac{3}{\Lambda}}$, we can embed the metric

$$ds^2 = -c^2 dt^2 + R^2 \sinh^2 \left(\frac{ct}{R} \right) (d\chi^2 + \sinh^2 \chi d\theta^2 + \sinh^2 \chi \sin^2 \theta d\phi^2)$$

using the parametrization of a hyperboloid.

$$\begin{cases} x_0 = R \sinh \left(\frac{ct}{R} \right) \cosh \chi \\ x_1 = R \cosh \left(\frac{ct}{R} \right) \\ x_2 = R \sinh \left(\frac{ct}{R} \right) \sinh \chi \cos \theta \\ x_3 = R \sinh \left(\frac{ct}{R} \right) \sinh \chi \sin \theta \cos \phi \\ x_4 = R \sinh \left(\frac{ct}{R} \right) \sinh \chi \sin \theta \sin \phi \end{cases} \Rightarrow -x_0^2 + \sum_{i=1}^4 x_i^2 = R^2$$

The three cases all satisfy the second Friedmann equation

$$\frac{\ddot{a}}{a} = \frac{1}{3} \Lambda c^2$$

and lead to a positive curvature using the convention $(-, +, +, +)$.

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \mathcal{R} + g_{\mu\nu} \Lambda = 0 \Rightarrow \mathcal{R} = 4\Lambda = \frac{12}{R^2}$$

Also, it is clear that the positive cosmological constant causes the universe expansion to accelerate outward. Even though our universe has matter $\rho \neq 0$, the de Sitter universe is still a useful model. When $a(t)$ becomes larger, the matter $\rho(a) \propto a^{-3}$ will become smaller but Λ remains the same.

Anti-de Sitter universe

With the same assumption $\rho = p = 0$ but a negative cosmological constant $\Lambda < 0$, the equation

$$\dot{a}^2 = \frac{\Lambda c^2}{3} a^2 - kc^2$$

has real solutions only for open universe $k = -1$.

$$a(t) = \sqrt{-\frac{3}{\Lambda}} \sin \left(\sqrt{-\frac{\Lambda c^2}{3}} t \right)$$

Calling $R = \sqrt{\frac{3}{\Lambda}}$, we can embed the metric

$$ds^2 = -c^2 dt^2 + R^2 \sin^2 \left(\frac{ct}{R} \right) (d\chi^2 + \sin^2 \chi d\theta^2 + \sin^2 \chi \sin^2 \theta d\phi^2)$$

using the parametrization of a hyperboloid.

$$\begin{cases} x_0 = R \cos \left(\frac{ct}{R} \right) \cosh \chi \\ x_1 = R \sin \left(\frac{ct}{R} \right) \\ x_2 = R \cos \left(\frac{ct}{R} \right) \sinh \chi \cos \theta \\ x_3 = R \cos \left(\frac{ct}{R} \right) \sinh \chi \sin \theta \cos \phi \\ x_4 = R \cos \left(\frac{ct}{R} \right) \sinh \chi \sin \theta \sin \phi \end{cases} \Rightarrow -x_0^2 - x_1^2 + \sum_{i=2}^4 x_i^2 = -R^2$$

The sinusoidal solution shows that the negative cosmological constant Λ causes the universe to accelerate inward, which leads to the Big Crunch, when $a(t)$ falls back to zero. Also, in $(-, +, +, +)$ convention, the metric gives a negative curvature.

$$\mathcal{R} = 4\Lambda = -\frac{12}{R^2}$$

In conclusion, we have learned 3-space slices in a 4D space.

$$S_3 : x_0^2 + x_1^2 + x_2^2 + x_3^2 = R^2$$

$$H_3 : -x_0^2 + x_1^2 + x_2^2 + x_3^2 = -R^2$$

In this section, we learned 4-hyperbolic slices in a 5D space.

$$dS_4 : -x_0^2 + x_1^2 + x_2^2 + x_3^2 + x_4^2 = R^2$$

$$AdS_4 : -x_0^2 - x_1^2 + x_2^2 + x_3^2 + x_4^2 = -R^2$$

8 Cosmological models

When we discuss a flat $k = 0$, closed $k = 1$, or open $k = -1$ universe, we define ρ_c as a crossover density between $k = 1$ and $k = -1$, called the critical density. Assuming $\Lambda = k = 0$, then the critical density ρ_c should satisfy

$$\frac{\dot{a}^2}{a^2} = H^2 = \frac{8\pi G}{3}\rho_c \quad \Rightarrow \quad \rho_c = \frac{3H^2}{8\pi G}$$

Then, we can alternatively write the 1st Friedmann equation as

$$\begin{aligned} \frac{\dot{a}^2 + kc^2}{a^2} &= \frac{8\pi G\rho + \Lambda c^2}{3} \quad \Rightarrow \quad H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G\rho}{3} + \frac{\Lambda c^2}{3} - \frac{kc^2}{a^2} \\ \Rightarrow \quad 1 &= \frac{8\pi G\rho}{3H^2} + \frac{\Lambda c^2}{3H^2} - \frac{kc^2}{a^2 H^2} \quad \Rightarrow \quad \Omega_{M/R} + \Omega_{\Lambda} + \Omega_k \end{aligned}$$

where

$$\Omega_{M/R} = \frac{\rho}{\rho_c} = \frac{8\pi G\rho}{3H^2}, \quad \Omega_{\Lambda} = \frac{\Lambda c^2}{3H^2}, \quad \Omega_k = -\frac{kc^2}{a^2 H^2}$$

are the density parameter (mass or radiation), the dark energy parameter, and the the curvature parameter, respectively.

Model with $\Lambda = 0$

If we consider $\Lambda = 0$, the 1st Friedmann equation becomes

$$\frac{kc^2}{a^2} = \Omega_{M/R} - 1$$

This tells us that

- When $\rho > \rho_c$, which means $\Omega_{M/R} > 1$, the universe is open $k = 1$.
- When $\rho = \rho_c$, which means $\Omega_{M/R} = 1$, the universe is flat $k = 0$.
- When $\rho < \rho_c$, which means $\Omega_{M/R} < 1$, the universe is closed $k = -1$.

The 2nd Friedmann equation tells us the universe expansion will always be decelerating.

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) \leq 0$$

Checking the 1st Friedmann equation again

$$\dot{a}^2 = \frac{8\pi G}{3}\rho a^2 - kc^2$$

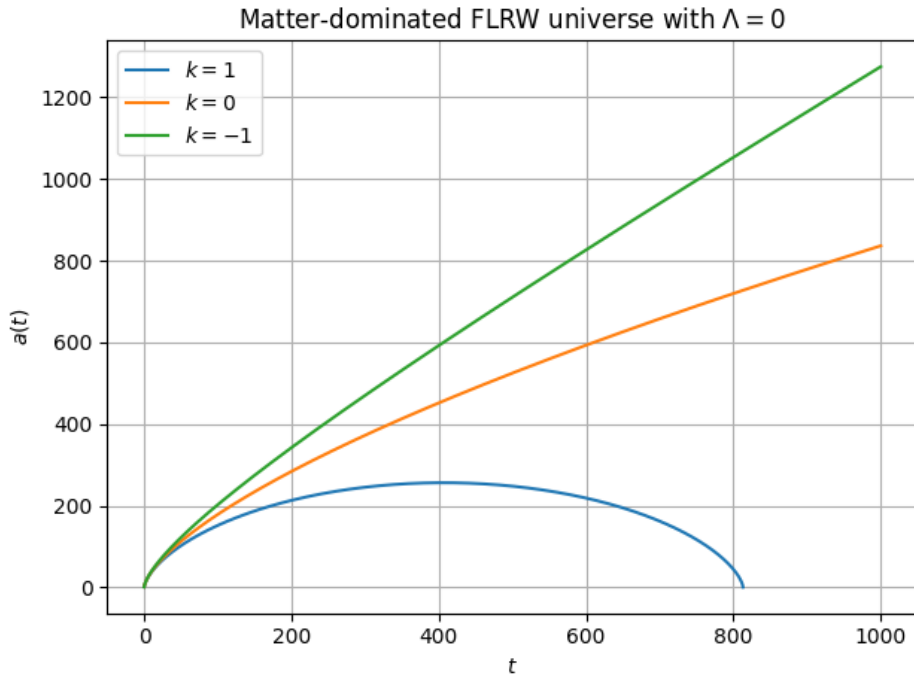
We can see the term $\rho a^2 \rightarrow 0$ as $a(t) \rightarrow \infty$. This gives us the asymptotic behavior of $a(t)$.

- For an open universe $k = -1$, we see $a(t)$ keeps increasing and $\dot{a} \rightarrow c$ as $a \rightarrow \infty$.
- For a close universe $k = 0$, we see $a(t)$ keeps increasing and $\dot{a} \rightarrow 0$ as $a \rightarrow \infty$
- For a close universe $k = 1$, we see $a(t)$ increases until $\dot{a} = 0$ at a finite scale

$$0 = \frac{8\pi G}{3} \left(\frac{\rho_0 a_0^{3(1+w)}}{a_{\max}^{3(1+w)}} \right) a_{\max}^2 - c^2 \quad \Rightarrow \quad a_{\max} = \left(\frac{8\pi G \rho_0}{3c^2} \right)^{\frac{1}{1+w}} a_0^3$$

and after that, $a(t)$ decreases.

Here is a simulation for a matter dominated universe with $\Lambda = 0$ and $k = 1, 0, -1$.



Model with $\Lambda \neq 0$

If we consider $\Lambda \neq 0$, the 1st Friedmann equation becomes

$$\frac{kc^2}{a^2} = \Omega_{M/R} + \Omega_{\Lambda} - 1$$

This tells us that

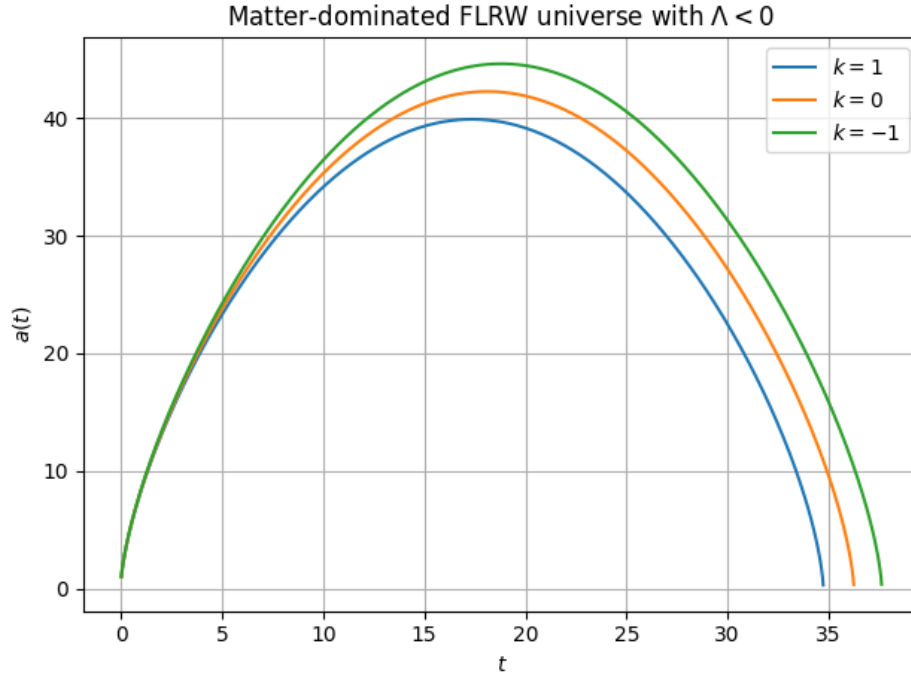
- When $\Omega_{M/R} + \Omega_{\Lambda} > 1$, the universe is open $k = 1$.
- When $\Omega_{M/R} + \Omega_{\Lambda} = 1$, the universe is flat $k = 0$.
- When $\Omega_{M/R} + \Omega_{\Lambda} < 1$, the universe is closed $k = -1$.

Case $\Lambda < 0$

The 2nd Friedmann equation tells us the universe expansion will always be decelerating.

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{1}{3}\Lambda c^2 < 0, \quad k = 1, 0, -1$$

Here is a simulation for a matter dominated universe with $\Lambda < 0$ and $k = 1, 0, -1$.

**Case $\Lambda > 0$**

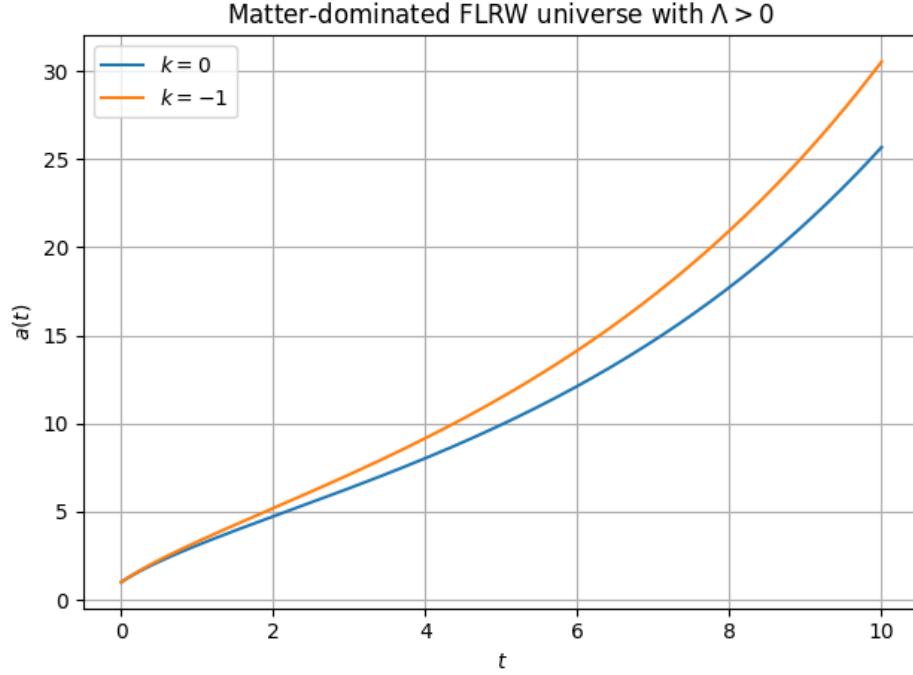
For $k = 0, -1$, the 1st Friedmann equation tells us the universe will always be expanding.

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho + \frac{1}{3}\Lambda c^2 - \frac{kc^2}{a^2} \geq \frac{1}{3}\Lambda c^2 > 0$$

Since the inequality is always satisfied, we see that $\dot{a}$ never changes sign. Also, from the 2nd Friedmann equation,

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{1}{3}\Lambda c^2$$

we see the universe expansion decelerates when ρ and p dominates the equation, but accelerates later when a is large causing Λ to dominate instead.



For $k = 1$, the initial condition determines whether the universe will always be expanding. Assuming the matter dominated universe has the density $\rho(a) = \rho_0 \left(\frac{a_0}{a}\right)^3$, we can solve for some $\rho_0 = \rho_*$, the universe expands to a finite scale a_* as $t \rightarrow \infty$ using the 1st and 2nd Friedmann equations.

$$\ddot{a} = 0 \quad \Rightarrow \quad 0 = \frac{\ddot{a}}{a_*} = -\frac{4\pi G}{3}\rho + \frac{1}{3}\Lambda c^2$$

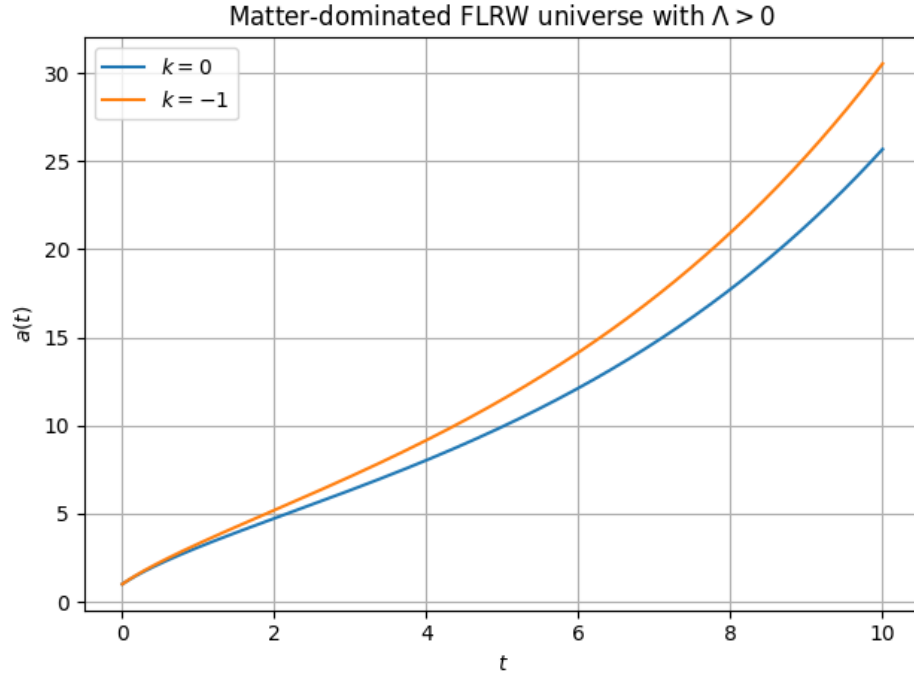
$$\dot{a} = 0 \quad \Rightarrow \quad 0 = \frac{\dot{a}^2}{a_*^2} = \frac{8\pi G}{3}\rho + \frac{1}{3}\Lambda c^2 - \frac{c^2}{a_*^2} = \Lambda c^2 - \frac{c^2}{a_*^2} \quad \Rightarrow \quad a_* = \frac{1}{\sqrt{\Lambda}}$$

Substitute a_* back to find ρ_*

$$0 = -\frac{4\pi G}{3}\rho + \frac{1}{3}\Lambda c^2 = -\frac{4\pi G}{3}\rho_* a_0^3 \Lambda^{\frac{3}{2}} + \frac{1}{3}\Lambda c^2 \quad \Rightarrow \quad \rho_* = \frac{c^2}{4\pi G a_0^3 \sqrt{\Lambda}}$$

- If $\rho_0 > \rho_*$, the expansion rate $\dot{a}$ never touches zero and then change sign, so the universe always expands.
- If $\rho_0 < \rho_*$, the expansion rate $\dot{a}$ touches zero at some scale $a_{\max} < a_*$ and then change sign, so the universe expands and then shrinks.

Here is a simulation for a matter dominated universe with $\Lambda < 0$ and $k = 1$ starting from $\rho_0 > \rho_*$, $\rho_0 = \rho_*$, and $\rho_0 < \rho_*$.



In the end of the theories, the estimated parameters are given below.

$$\begin{cases} \Omega_k \approx 0 & \Rightarrow & k \approx 0 \\ \Omega_{M/R} \approx 0.32 & \Rightarrow & \rho \approx 2.73 \times 10^{-27} \text{ kg} \cdot \text{m}^{-3} \\ \Omega_\Lambda \approx 0.68 & \Rightarrow & \Lambda \approx 1.08 \times 10^{-52} \text{ m}^{-2} \end{cases}$$